

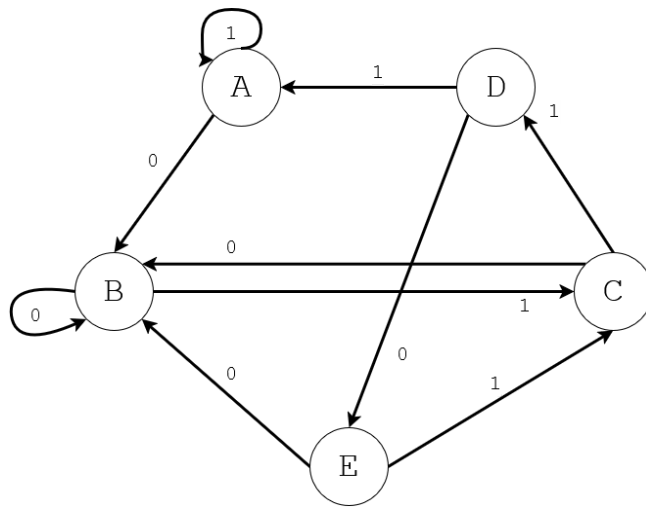
#PSDisFun 

1. Specification

0110 sequence recognizer, with D-Flip Flop, Moore model.

2. Formulation

State Diagram:



State Table:

Present State	Next State		Output
	X = 0	X = 1	
A	B	A	0
B	B	C	0
C	B	D	0
D	E	A	0
E	B	C	1

3. State Assignment

Dengan *binary counting order assignment*, definisikan

A	B	C	D	E
000	001	010	011	100

4. Flip-Flop Input Equation Determination

Diperlukan tiga buah D Flip-Flop yang dinamakan d2, d1, dan d0.

Present State			Next State						Output
			X = 0			X = 1			
y2	y1	y0	d2	d1	d0	d2	d1	d0	z
0	0	0	0	0	1	0	0	0	0
0	0	1	0	0	1	0	1	0	0
0	1	0	0	0	1	0	1	1	0
0	1	1	1	0	0	0	0	0	0
1	0	0	0	0	1	0	1	0	1

5. Output Equation Determination

Dengan menggunakan K-Map, diperoleh

*note: A = y2, B = y1, C = y0, D = X (input)

d2	d1	d0	z
$\begin{array}{c cccc} & \bar{C}\bar{D} & \bar{C}D & CD & C\bar{D} \\ \hline \bar{A}\bar{B} & 0 & 0 & 0 & 0 \\ \bar{A}B & 0 & 0 & 0 & 1 \\ AB & x & x & x & x \\ A\bar{B} & 0 & 0 & x & x \end{array}$	$\begin{array}{c cccc} & \bar{C}\bar{D} & \bar{C}D & CD & C\bar{D} \\ \hline \bar{A}\bar{B} & 0 & 0 & 1 & 0 \\ \bar{A}B & 0 & 1 & 0 & 0 \\ AB & x & x & x & x \\ A\bar{B} & 0 & 1 & x & x \end{array}$	$\begin{array}{c cccc} & \bar{C}\bar{D} & \bar{C}D & CD & C\bar{D} \\ \hline \bar{A}\bar{B} & 1 & 0 & 0 & 1 \\ \bar{A}B & 1 & 1 & 0 & 0 \\ AB & x & x & x & x \\ A\bar{B} & 1 & 0 & x & x \end{array}$	$\begin{array}{c cccc} & \bar{C}\bar{D} & \bar{C}D & CD & C\bar{D} \\ \hline \bar{A}\bar{B} & 0 & 0 & 0 & 0 \\ \bar{A}B & 0 & 0 & 0 & 0 \\ AB & x & x & x & x \\ A\bar{B} & 1 & 1 & x & x \end{array}$

d2	$y1.y0.X'$
d1	$y2.X + y1'.y0.X + y1.y0'.X$
d0	$y1'.X' + y1.y0'$
z	$y2$

6. Technology Mapping

